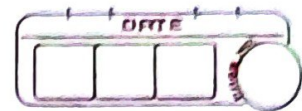


Digital Logic Design

Module 1 [1.1]



Nibble - 4

Byte - 8

Word - 16

Quad - 32

- # Number Systems $\Rightarrow$
- Decimal [10] $\Rightarrow$ 0-9
 - Binary [2] $\Rightarrow$ 0-1
 - Octal [8] $\Rightarrow$ 0-7
 - Hexadecimal [16] $\Rightarrow$ 0-9 & A-F

Converting Decimal to other system

1) To binary

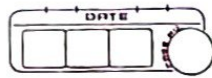
$$\rightarrow (27.315)_{10} = (11011.0101)_2$$

2	27		$0.315 \times 2 = 0.63$	0
2	13 - 1		$0.63 \times 2 = 1.26$	1
2	6 - 1		$0.26 \times 2 = 0.52$	0
2	3 - 0		$0.52 \times 2 = 1.04$	1
2	1 - 1			
	0 - 1			

2) To Octal

$$\rightarrow (275.4)_{10} = (423.3146)_8$$

8	275		$0.4 \times 8 = 3.2$	3
8	34 - 3		$0.2 \times 8 = 1.6$	1
8	4 - 2		$0.6 \times 8 = 4.8$	4
	0 - 4		$0.8 \times 8 = 6.4$	6



3) # To Hexadecimal

$$\rightarrow (382.525)_{10} = (17E.8C)_{16}$$

16	382
16	23 - 14 (E) ↑
16	1 - 7
	0 - 1

$$0.525 \times 16 = 8.4 - 8 \downarrow$$

$$0.4 \times 16 = 6.4 - 6 \downarrow$$

Binary to Other System

1) To Decimal

$$\rightarrow (11010.11)_2 = (26.75)_{10}$$

$$\begin{matrix} (2^4) & (2^3) & (2^2) & (2^1) & (2^0) & (2^{-1}) & (2^{-2}) \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 \end{matrix}$$

$$= (16 + 8 + 0 + 2 + 0) + (0.5 + 0.25)$$

$$= (26.75)_{10}$$

~~Note → To Octal & Hexadecimal~~



Note ⇒ For (Octal & Hexadecimal) $\longleftrightarrow$ Binary

0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10 (A)	1	0	1	0
11 (B)	1	0	1	1
12 (C)	1	1	0	0
13 (D)	1	1	0	1
14 (E)	1	1	1	0
15 (F)	1	1	1	1

• For Octal ⇒ Make pair of 3

• For Hexadecimal ⇒ Pair of 4

2) To Octal ⇒ $(111011.10)_2$

$$= (\underline{111} \underline{011} \underline{.100})_2 = (73.4)_8$$

3) To Hexadecimal ⇒ 111110011.011

$$= \underline{00011111} \underline{0011} \underline{.0110}$$

$$= (1F3E)_{16}$$

Octal to Other System

1) To Binary $\Rightarrow (57)_8 = (101111)_2$

$(7245)_8 = (111010 \cdot 100101)_2$

2) To Decimal $\Rightarrow (26.24)_8$

$\rightarrow (2 \times 8^1) + (6 \times 8^0) + (2 \times 8^{-1}) + (4 \times 8^{-2})$

$= (22.3125)_{10}$

3) To Hexadecimal $\Rightarrow (1267.34)_8$

$\rightarrow (1267.34)_8 = (001010110111 \cdot 01110000)_2$

$= (2B7.7)_{16}$

Hexadecimal to Other System

1) Hexadecimal to Binary

$(3C5F)_{16} = (10110110 \cdot 01011111)_2$

$(DAB.C)_{16} = (110110101011 \cdot 1100)_2$

2) To Octal

$\rightarrow (FACE.25)_{16} = (001111101011001110 \cdot 001001010)_2$
 $= (175316.112)_8$

3) To Decimal

$\rightarrow (1AG)_{16} = (16^2 \times 1) + (16^1 \times 10) + (16^0 \times 6)$
 $= (422)_{10}$

$$(-2^{n-1} + 1) \longrightarrow (+2^{n-1} + 1)$$

→ 1's complement $\Rightarrow$ • Represent negative of a no.

• Invert bits

$$\begin{aligned} \Rightarrow (5)_{10} &= (0101)_2 \\ (-5)_{10} &= (\underline{1010})_2 \end{aligned}$$

→ 2's complement $\Rightarrow$ • 1's complement + 1

$$(-2^n \longrightarrow 2^{n-1} - 1)$$

$$\Rightarrow (5)_{10} = (0101)_{2^1}$$

$$\begin{aligned} (-5)_{10} &= (1010 + 1)_{2^1} \\ &= (1011)_{(2, 2^1)} \end{aligned}$$

Binary Addition

a	b	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Note $(1 + 1 + 1)$ 1 1

Binary Subtraction

1) Borrow Method

a	b	Subtract	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$\begin{array}{r} \text{Question} \\ 1011011 \\ - 0010010 \\ \hline 1001001 \end{array}$$

$$\begin{array}{r} \text{Question} \\ 1010110 \\ - 0101010 \\ \hline 0101100 \end{array}$$

$$\begin{array}{r} \text{Question} \\ \begin{matrix} (1) & (1) & (1) & (2) & (2) \\ 1010110 \\ - 00111010 \\ \hline 10011100 \end{matrix} \end{array}$$



2) Using 1's Complement

Question $6 - 2 = ?$

Answer $6 - 2 = 6 + (-2)$

$$6 = 0110$$

$$2 = 0010$$

$$(-2) = 1101$$

$$\begin{array}{r} \text{Step 1} \Rightarrow 6 = 0110 \\ + (-2) = 1101 \\ \hline 10011 \\ \text{Carry} \rightarrow \boxed{\text{Represent +ve no}} \end{array}$$

Step 2 $\Rightarrow$ Add that carry

$$\begin{array}{r} 6 + (-2) = 0011 \\ + 1 \\ \hline 0100 = 4 \text{ Answer} \end{array}$$

Question $2 - 6 = 2 + (-6)$

Solution $2 = 0010$

$$6 = 0110$$

$$(-6) = 1001$$

2) Now here take 1's complement

$$\begin{array}{l} \rightarrow 1011 \rightarrow 0100 \\ = -4 \end{array}$$

$$\begin{array}{r} 1) \rightarrow 2 - 6 = 0010 \\ + 1001 \\ \hline 1011 \end{array}$$

[No Carry $\rightarrow$ -ve]



3) Using 2's complement

Question $6 - 2 = ?$

Soln $\Rightarrow 6 = 0110$

$$2 = 0010$$

$$(-2) = \begin{pmatrix} 1101 \\ + 1 \end{pmatrix} = (1110)_{2's}$$

$$\rightarrow 6 - 2 = 6 + (-2) = 0110$$

$$+ 1110$$

$$\hline 10100$$

Carry represent $\rightarrow$ answer = 4

Question $2 - 6 = ?$

Soln $\Rightarrow 2 = 0010$

$$6 = 0110$$

$$(-6) = \begin{pmatrix} 1001 \\ + 1 \end{pmatrix} = (1010)_{2's}$$

$$\rightarrow 2 - 6 = 2 + (-6) = 0010$$

$$+ 1010$$

$$\hline 1100$$

$\rightarrow$ No carry = -ve (Again take 1's complement)

$$1100 \rightarrow 0011$$

$$+ 1$$

$$\hline 0100 \rightarrow 4 \rightarrow -4 \text{ (No carry)}$$

Binary Coded Decimal [BCD / 8421 code]

→ Represent each digit of decimal no. separately by binary & club it [Binary of 4 bits]

→ 72 → 0111 0010

• 56.13 → 0101 0110 . 0001 0011

BCD Addition

Question 7 0111
 + 2 0010
 —————
 9 1001 → 9

Fine if
 $\text{sum} \leq 9$ &
 $\text{carry} = 0$

Question 7 0111
 + 5 0101
 —————
 12 1100 → > 9 (Add 6)

+ 0110
 0001 0010
 (1) (2) → 12

S.E. [10, 15]
 add 6 to
 sum

If we get
 carry & $\text{sum} \leq 9$
 add 6 to previous
 bit

Question 28 0010 1000
 + 39 0011 1001
 —————
 67 0110 0001

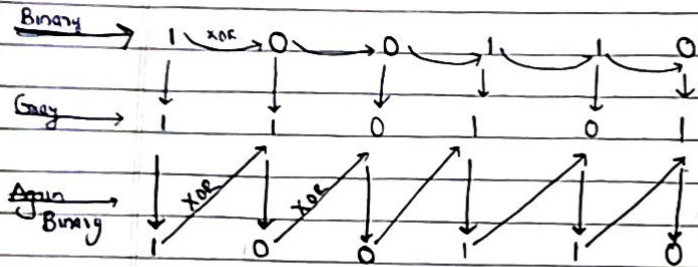
+ 0110
 0110 0111
 7

Gray Code

→ Need XOR [Exclusive OR]

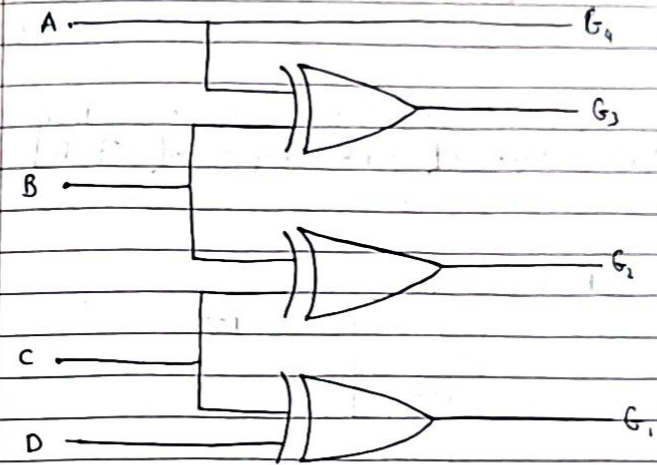
$$0 \oplus 0 = 1 \oplus 1 = 0$$

$$0 \oplus 1 = 1 \oplus 0 = 1$$



Number	Binary	Gray
0	0 0 0	0 0 0
1	0 0 1	0 0 1
2	0 1 0	0 1 1
3	0 1 1	0 1 0
4	1 0 0	1 1 0
5	1 0 1	1 1 1
6	1 1 0	1 0 1
7	1 1 1	1 0 0

Binary to Gray



Need ⇒ Coding & Decoding
 Error Detection & Correction
 Efficient Transmission

Excess 3 code [BCD code]

→ Here, basically add 3 (0011) to the BCD code [Each 4 digit of each byte of BCD]

Example → 5 → BCD = 0101
 E3C = $\begin{array}{r} 0101 \\ + 0011 \\ \hline 1000 \end{array}$ = 1000

2) 72 → 0111 0010 = BCD

E3C = $\begin{array}{r} 0111 \quad 0010 \\ 0011 \quad 0011 \\ \hline 1010 \quad 0101 \end{array}$

3) 23.15 → 0010 0011 . 0001 0101
 [BCD → E3C] $\begin{array}{r} + 0011 \quad 0011 \quad 0011 \quad 0011 \\ \hline 0101 \quad 0110 \quad 0100 \quad 1000 \end{array}$

[E3C → BCD] $\begin{array}{r} 0101 \quad 0110 \quad 0100 \quad 1000 \\ - 0011 \quad 0011 \quad 0011 \quad 0011 \\ \hline 0010 \quad 0011 \quad 0001 \quad 0101 \end{array}$

→ Addition

1) 37 + 28 = 65

→ $\begin{array}{r} 37 = 0011 \quad 0111 \\ + 33 = 0011 \quad 0011 \\ \hline 6 \quad 10(n) \quad 0110 \quad 1010 \\ (6) \quad (10) \end{array}$ $\begin{array}{r} 28 = 0010 \quad 1000 \\ + 33 = 0011 \quad 0011 \\ \hline 5 \quad 11(8) \quad 0101 \quad 1011 \\ 5 \quad (11) \end{array}$

⇒ $\begin{array}{r} 37 \Rightarrow 0111 \quad 0110 \\ + 28 \Rightarrow 0101 \quad 1011 \\ \hline 65 \Rightarrow 1100 \quad 0101 \\ + 33 \Rightarrow 0011 \quad 1000 \\ \hline 98 \Rightarrow 1001 \quad 1000 \\ (9) \quad (8) \end{array}$

If carry → add 3

No carry → minus 3

2) $\begin{array}{l} \nearrow \text{1 digit} \\ 7 + 3 = 10 \end{array}$ $\begin{array}{l} \searrow \text{2 digit} \end{array}$

⇒ $\boxed{07 + 03 = 10}$

⇒ $\begin{array}{r} 07 = 0011 \quad 0110 \\ + 03 = 0011 \quad 0110 \\ \hline 10 = 0111 \quad 0000 \\ + 33 = 0100 \quad 0011 \\ \hline 43 = 1011 \quad 0101 \\ (4) \quad (3) \end{array}$

07 → $\begin{array}{r} 0000 \quad 0111 \\ 0011 \quad 0011 \\ \hline 0011 \quad 1010 \end{array}$

03 → $\begin{array}{r} 0000 \quad 0011 \\ 0011 \quad 0011 \\ \hline 0011 \quad 0110 \end{array}$

Parity & Error Detection

→ Used for Error Detection

→ Even parity → No of 1's = even

Odd parity → No of 1's = odd

→ Add 1 bit to make it as per your parity

→ Only single bit error detection can be detected

Rule 2

rg

x.

x+

tion Law ⇒